

Operating Systems Laboratory - Assignment 1

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1 Introduction

This assignment involved writing a C++ application to sharpen images in PPM format. The pipeline consists of three stages – `S1_smoothen()`, `S2_find_details()`, and `S3_sharpen()` – along with file read and file write. Execution times for all five stages were measured across seven sample images of increasing size, each run averaged over 5 repetitions using the C++ `chrono` library.

2 Methodology

Each image was processed 5 times, and the time taken for file read, `S1`, `S2`, `S3`, and file write was recorded in each run. The reported values are the averages of these 5 runs, measured in milliseconds.

3 Functions

- `S1_smoothen()`: Applies a 3x3 averaging filter over each pixel's neighborhood to blur the image and reduce noise.
- `S2_find_details()`: Subtracts the smoothed image from the original to extract edge/detail information, clipping negative values to zero.
- `S3_sharpen()`: Adds the extracted details back to the original image, clipping values above 255, to produce the sharpened output.

4 Results

Image	Read	S1	S2	S3	Write
1.ppm	5.99	13.63	4.39	4.90	2.28
2.ppm	19.70	43.53	10.37	11.38	5.47
3.ppm	23.80	67.60	18.99	20.97	10.93
4.ppm	46.70	132.63	37.37	39.10	19.98
5.ppm	60.92	175.82	47.44	51.40	28.93
6.ppm	131.92	398.10	106.72	116.05	58.34
7.ppm	362.66	1087.02	293.95	313.81	160.27

Table 1: Average execution time (ms) for each stage, per image

5 Discussion

- Execution time increases with image size across all five stages, as expected since more pixels must be processed.
- `S1.smoothen` takes the longest in every image, since it computes a 3x3 neighborhood average for every pixel, giving it a higher constant factor than the other stages.
- `S2.find_details` and `S3.sharpen` take comparable time to each other, as both perform a single pixel-wise arithmetic operation (subtraction and addition, respectively) with clipping.
- File read consistently takes longer than file write, likely due to the overhead of parsing the PPM header and pixel values on input, whereas output is a more direct write of already-formatted data.